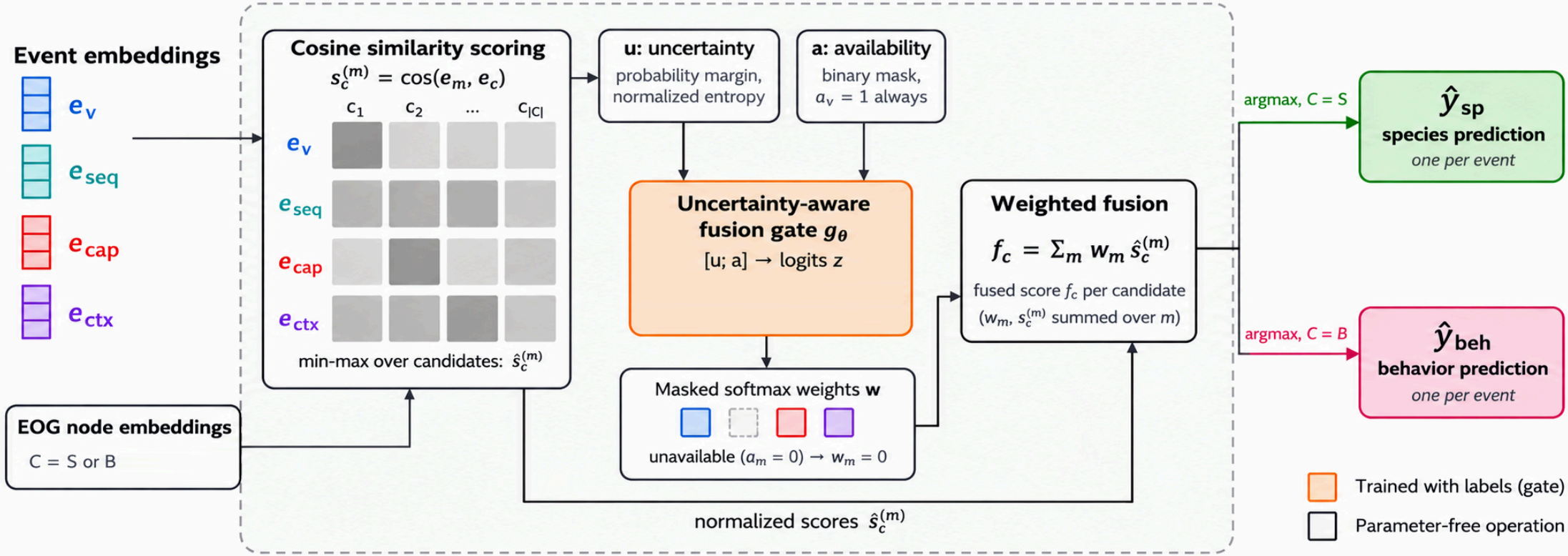

Event embeddings
e_v
e_{seq}
e_{cap}
e_{ctx}
EOG node embeddings
C = S or B
Cosine similarity scoring
$s_c^{(m)} = \cos(e_m, e_c)$
c_1 c_2 ... $c_{|C|}$
e_v
e_{seq}
e_{cap}
e_{ctx}
min-max over candidates: $\hat{s}_c^{(m)}$
u: uncertainty
probability margin, normalized entropy
a: availability
binary mask, $a_v = 1$ always
Uncertainty-aware fusion gate g_θ
[u; a] → logits z
Masked softmax weights w
unavailable ($a_m = 0$) → $w_m = 0$
Weighted fusion
$f_c = \Sigma_m w_m \hat{s}_c^{(m)}$
fused score f_c per candidate
(w_m, $s_c^{(m)}$ summed over m)
normalized scores $\hat{s}_c^{(m)}$
argmax, C = S
$\hat{y}_{sp}$
species prediction
one per event
argmax, C = B
$\hat{y}_{beh}$
behavior prediction
one per event
Trained with labels (gate)
Parameter-free operation